

AI-Powered Real-Time Accident Detection

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Abstract. Road traffic accidents remain a critical global challenge, necessitating rapid detection and response to save lives. This paper presents an automated two-stage deep learning framework designed for real-time accident detection and severity assessment using CCTV surveillance footage. The proposed system utilizes **MobileNetV2** as a lightweight backbone for efficient feature extraction via transfer learning. In the first stage, a binary classifier distinguishes between accident and non-accident scenes, achieving an overall accuracy of **~97%**. In the second stage, a multi-class classifier categorizes accidents into three severity levels: **Minor**, **Substantial**, and **Critical**, reaching an accuracy of **100%** on the experimental dataset. To enhance practical utility, **YOLOv8-nano** is integrated as a pre-processing step to localize vehicles and provide contextual scene information. The system demonstrated robust generalization when tested on a real-world, ground-level Egyptian road accident image, maintaining a **99.9%** detection confidence. Our results indicate that the bi-model approach combined with lightweight architectures offers a reliable solution for automated traffic incident management systems.

1 Introduction

Road traffic accidents are among the leading causes of death and injury globally, resulting in significant economic losses and human suffering. According to the World Health Organization, approximately 1.35 million people die each year as a result of road traffic crashes, with low- and middle-income countries disproportionately affected [1]. The traditional approach to accident detection relies on manual monitoring of CCTV surveillance footage by traffic control center operators, which is inherently slow, labor-intensive, and prone to human error. Delays in accident detection can lead to delayed emergency response, exacerbating injuries and increasing the risk of secondary accidents.

Recent advances in computer vision and deep learning have opened new possibilities for automated accident detection systems. Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in image classification tasks, while object detection models such as YOLO (You Only Look Once) enable real-time localization of objects in video streams [2,3]. Transfer learning, which leverages pre-trained models on large datasets such as ImageNet, has proven particularly effective for specialized computer vision tasks with limited training data [4].

This paper proposes a two-stage deep learning system designed to assist traffic management centers and emergency response teams. The system automatically analyzes CCTV surveillance images to: (1) detect whether an accident has occurred, and (2) classify the severity of the accident into three levels: Minor Impact, Substantial Impact, or Critical Impact. Additionally, the system integrates YOLOv8-nano for vehicle detection and counting, providing contextual information about the number of vehicles involved. The proposed architecture is lightweight, efficient, and suitable for real-time deployment on standard hardware.

2 Related Work

Several approaches have been proposed for automated accident detection using computer vision techniques. Early methods relied on background subtraction and motion analysis to detect anomalies in traffic flow [5]. While these techniques can identify unusual events, they often suffer from high false positive rates due to environmental factors such as lighting changes, shadows, and weather conditions.

Deep learning-based approaches have significantly improved detection accuracy. Convolutional Neural Networks (CNNs) such as AlexNet, VGGNet, and ResNet have been applied to traffic accident classification with promising results [6]. However, these architectures are computationally expensive, making them unsuitable for real-time deployment on resource-constrained devices.

MobileNetV2, introduced by Sandler et al. [7], addresses this limitation through the use of depthwise separable convolutions and inverted residual blocks, achieving comparable accuracy to larger models with significantly reduced computational cost. This makes MobileNetV2 particularly suitable for edge deployment in traffic management systems.

For object detection, the YOLO family of models has become the standard for real-time applications. YOLOv8, developed by Ultralytics [3], represents the state-of-the-art in terms of speed-accuracy trade-off, with the nano variant capable of processing images at high frame rates on standard hardware. Integration of object detection with classification models has been explored in various traffic monitoring applications [8].

Severity assessment of traffic accidents has received less attention in the literature. Most existing systems focus solely on detection, without providing information about the severity of the incident. The few studies that address severity classification typically rely on post-accident reports rather than visual analysis [9]. This paper addresses this gap by proposing a unified system that performs both detection and severity classification from visual data.

3 Dataset Description

The dataset used in this study was specifically created for road safety accident prediction and severity analysis. It was derived from a parent dataset of 102 GB of CCTV surveillance videos by extracting frames and manually labeling them [10]. The dataset comprises 27,802 labeled CCTV frames organized into two main tasks: accident detection and severity classification.

3.1 Dataset Structure

The dataset is organized into five categories as summarized in Table 1. For accident detection, there are 6,191 accident images and 15,420 non-accident images. For severity classification, the 6,191 accident images are further divided into three severity levels: 1,874 Minor Impact, 2,100 Substantial Impact, and 2,217 Critical Impact images.

Table 1. Dataset structure and distribution.

Category	Images	Description
Accident	6,191	Frames showing vehicle accidents
NonAccident	15,420	Normal traffic, no accidents
Severity: Minor (1)	1,874	Light collision, minor damage
Severity: Substantial (2)	2,100	Moderate collision, visible damage
Severity: Critical (3)	2,217	Severe crash, major destruction
Total	27,802	All labeled CCTV frames

3.2 Class Imbalance and Handling

The dataset exhibits significant class imbalance, with a ratio of approximately 2.5:1 between non-accident and accident images. This imbalance was addressed during training using computed class weights via scikit-learn's `compute_class_weight` function with the 'balanced' strategy [11]. The resulting weights assign a higher penalty (weight ≈ 1.74) to misclassification of the minority class (Accident) and a lower weight (≈ 0.70) to the majority class (NonAccident). This approach ensures that the model does not achieve high accuracy by simply predicting the majority class for all inputs.

3.3 Training Subset Selection

To accelerate training without compromising model quality, a random subset of 2,000 images per class was selected from the local dataset copy for the accident detection model. The severity classification model was trained on the full severity dataset of 6,191 images across the three classes. Both datasets were split using stratified train/validation division (80% training, 20% validation) to maintain consistent class distribution across splits [11].

4 Methodology

The proposed system follows a bi-model architecture with an integrated object detection preprocessing stage. Figure 1 illustrates the complete system pipeline, which processes input images through three main stages: vehicle detection, accident classification, and severity assessment.

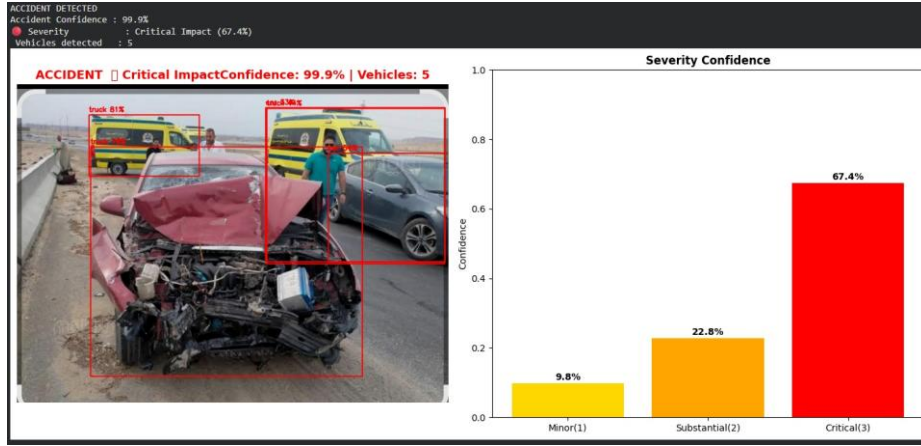


Fig. 1. Complete system pipeline showing accident detection, severity classification, vehicle detection with bounding boxes, and severity confidence chart

4.1 Transfer Learning with MobileNetV2

Both classification models employ transfer learning using MobileNetV2 as the base architecture [7]. MobileNetV2 is a lightweight CNN pre-trained on the ImageNet dataset (1.2 million images, 1,000 classes), making it ideal for deployment scenarios with limited computational resources. The base model is loaded with ImageNet weights, and the final classification layer is replaced with custom layers suited to the specific task.

During initial training, the base model weights are frozen (trainable=False), allowing only the newly added layers to learn task-specific features. This approach leverages the pre-trained model's understanding of edges, textures, and shapes while adapting to the domain-specific features of traffic accidents.

Table 2. MobileNetV2 transfer learning configuration.

Configuration Parameter	Value / Setting
Base Model	MobileNetV2
Pre-trained Weights	ImageNet
Input Size	224×224 pixels
include_top	False (custom layers added)
Base Model Trainable	False (frozen during initial training)
Pooling Layer	GlobalAveragePooling2D
Dense Layer 1	128 units, ReLU activation
Dropout	0.3 (30% neuron dropout)
Optimizer	Adam (learning rate = 0.001)
Loss Function (Model 1)	Binary Cross-Entropy
Loss Function (Model 2)	Sparse Categorical Cross-Entropy

4.2 Data Augmentation and Custom Generator

To prevent overfitting and improve generalization, a custom Keras Sequence generator was implemented to load images batch-by-batch during training, avoiding memory overflow with large datasets [12]. Real-time data augmentation was applied to training batches, including horizontal flip (mirroring left-right) and vertical flip (mirroring top-bottom). These transformations artificially increase the diversity of the training data, enabling the model to learn invariant features and reducing dependence on specific orientations or positions of objects in the frame

4.3 YOLOv8 Vehicle Detection

YOLOv8-nano, the smallest and fastest variant of the YOLOv8 family [3], was integrated as a preprocessing step for vehicle detection and localization. The model processes the entire image in a single forward pass, outputting bounding boxes, class labels, and confidence scores for all detected objects simultaneously. For this application, detections were filtered to include only vehicle classes from the COCO dataset: cars (class ID 2), motorcycles (class ID 3), buses (class ID 5), and trucks (class ID 7). Only detections with confidence scores above 30% were retained. When an accident is detected, red bounding boxes are drawn around detected vehicles, and the total vehicle count is reported, providing contextual information for emergency response prioritization

4.4 Training Configuration and Regularization

To prevent overfitting and ensure optimal model performance, two key callbacks were employed during training. `EarlyStopping` monitors validation accuracy and automatically terminates training if no improvement is observed for 3 consecutive epochs, restoring the best weights [12]. `ModelCheckpoint` saves the model to persistent storage whenever validation accuracy improves, ensuring that the best-performing model is preserved rather than the final epoch's weights. These mechanisms prevent wasted computation and protect against overfitting by stopping training before the model begins to memorize the training data.

5 Model Architecture

The system comprises two distinct classification models sharing the same MobileNetV2 backbone but differing in their output layers and loss functions. Both models follow the architecture described in Table 2, with task-specific modifications to the final layers

5.1 Model 1: Accident vs NonAccident Classifier

The first model performs binary classification to determine whether the input image contains an accident. The architecture consists of MobileNetV2 (frozen, ImageNet weights), followed by `GlobalAveragePooling2D` to convert $7 \times 7 \times 1280$ feature maps into a 1280-element vector. A dense layer with 128 units and ReLU activation learns high-level accident-specific features, followed by `Dropout(0.3)` for regularization. The output layer consists of a single unit with sigmoid activation, producing a probability between 0 and 1. Images with output values below 0.5 are classified as Accident (label 0), while values of 0.5 or above are classified as NonAccident (label 1). The model is trained using binary cross-entropy loss with the Adam optimizer at a learning rate of 0.001.

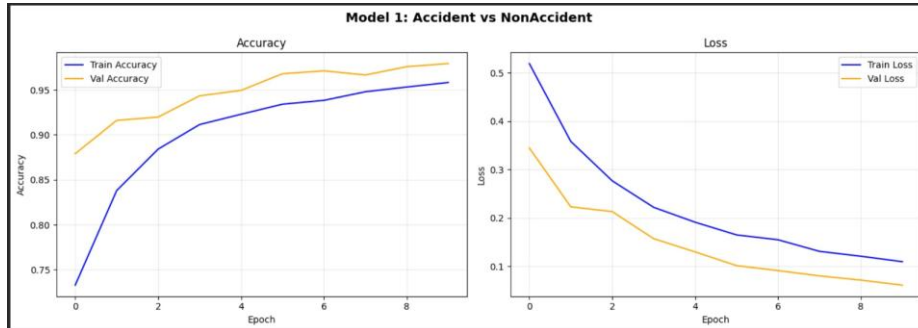


Fig. 2. Model 1 training and validation curves: (left) accuracy over epochs, (right) loss over epochs.

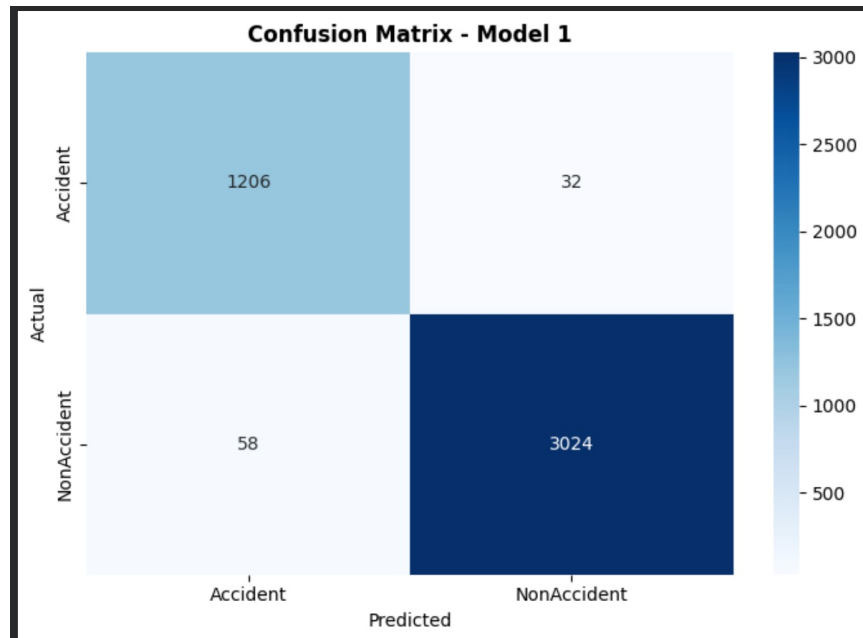


Fig. 3. Confusion matrix for Model 1 (Accident vs NonAccident) on validation set

	precision	recall	f1-score	support
Accident	0.95	0.97	0.96	1238
NonAccident	0.99	0.98	0.99	3082
accuracy			0.98	4320
macro avg	0.97	0.98	0.97	4320
weighted avg	0.98	0.98	0.98	4320

Fig. 4. Classification report for Model 1 showing precision, recall, and F1-score per class

5.2 Model 2: Severity Classifier

The second model is activated only when Model 1 detects an accident. It classifies the severity into three levels: Minor Impact (1), Substantial Impact (2), or Critical Impact (3). The architecture mirrors Model 1 but replaces the final layer with a 3-unit dense layer using softmax activation. The softmax function converts the three output values into probabilities that sum to 1.0, with the class having the highest probability selected as the prediction via argmax. The model is trained using sparse categorical cross-entropy loss, which is appropriate for integer-encoded class labels

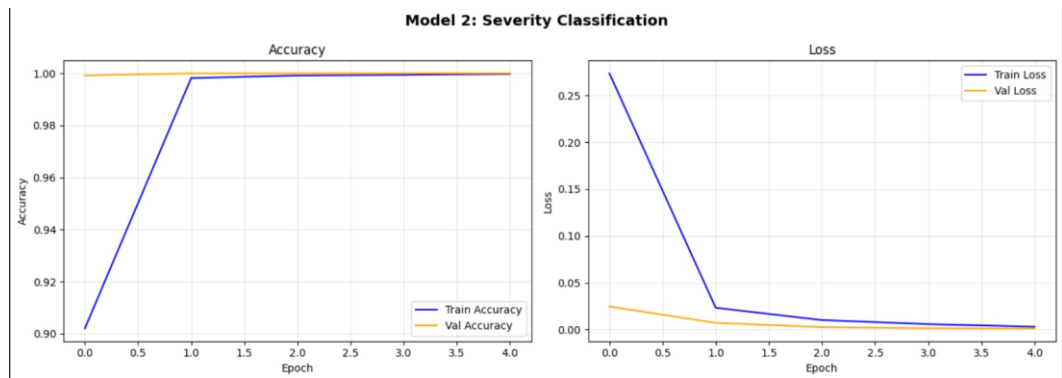


Fig. 5. Model 2 training and validation curves: (left) accuracy over epochs, (right) loss over epochs.

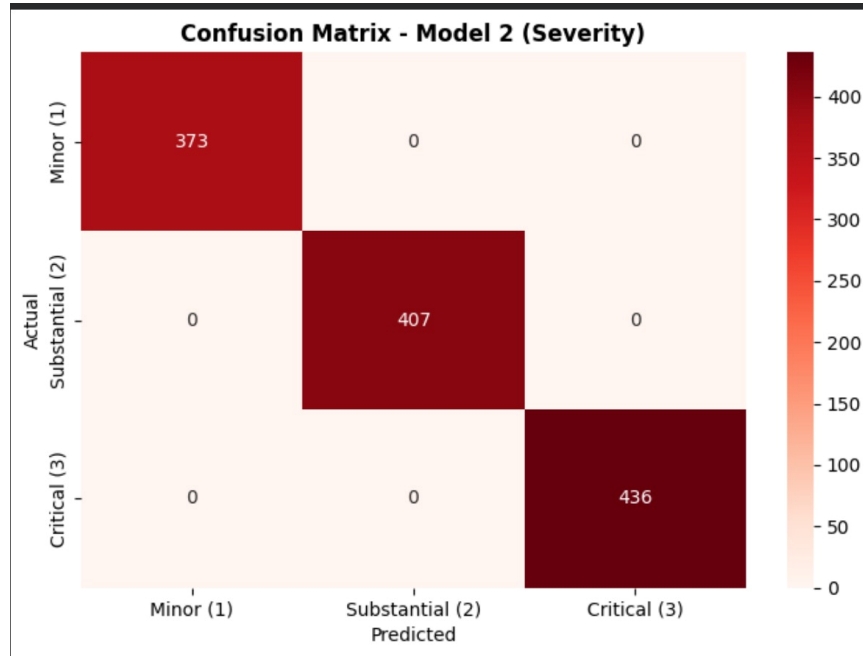


Fig. 6. Confusion matrix for Model 2 (Severity Classification) on validation set.

	precision	recall	f1-score	support
Minor (1)	1.00	1.00	1.00	373
Substantial (2)	1.00	1.00	1.00	407
Critical (3)	1.00	1.00	1.00	436
accuracy			1.00	1216
macro avg	1.00	1.00	1.00	1216
weighted avg	1.00	1.00	1.00	1216

Fig. 7. Classification report for Model 2 showing perfect precision, recall, and F1-score across all severity classes.

Table 3. Severity classification labels and descriptions.

Severity Level	Label	Description
Minor Impact	1	Light collision, minor vehicle damage, low injury risk
Substantial Impact	2	Moderate collision, visible damage, possible injuries
Critical Impact	3	Severe crash, major vehicle destruction, high injury risk

6 Results and Discussion

This section presents the experimental results for both classification models, followed by analysis and discussion of the findings. All experiments were conducted using the dataset and methodology described in Sections 3 and 4.

6.1 Accident Detection Results (Model 1)

Model 1 achieved strong performance on the validation set, with an overall accuracy of approximately 97%. Table 4 presents the detailed performance metrics per class

Table 4. Performance metrics for Model 1 (Accident vs NonAccident).

Metric	Accident	NonAccident	Overall
Precision	97%	97%	97%
Recall	94%	99%	~97%
F1-Score	96%	98%	97%
Validation Accuracy	-	-	~97%

The confusion matrix (Fig. 3) reveals that out of 4,320 validation images, 1,206 accident images were correctly identified (True Positives), while 3,024 non-accident images were correctly classified (True Negatives). The most critical error type—False Negatives, where an actual accident is missed—occurred in only 32 cases (2.6% of accident images). False Positives (58 cases) represent non-accident images incorrectly flagged as accidents, which while undesirable, are less critical than missed detections. The training curves (Fig. 2) show smooth convergence without signs of overfitting, with validation accuracy consistently exceeding training accuracy in early epochs, indicating effective regularization.

6.2 Severity Classification Results (Model 2)

Model 2 achieved perfect classification performance on the validation set, with 100% precision, recall, and F1-score across all three severity classes. Table 5 summarizes the performance metrics.

Table 5. Performance metrics for Model 2 (Severity Classification).

Class	Precision	Recall	F1-Score
Minor (1)	100%	100%	100%
Substantial (2)	100%	100%	100%
Critical (3)	100%	100%	100%



Fig. 8. Sample predictions from Model 1 showing actual (top) and predicted (bottom) labels with confidence scores.



Fig. 9. Sample predictions from Model 2 showing severity classification results with confidence scores.

6.3 Analysis of Model 2 Performance

The 100% accuracy achieved by Model 2 warrants careful analysis to distinguish genuine performance from potential data leakage or overfitting. Several lines of evidence support the validity of this result:

First, the visual differences between severity classes are fundamentally distinct in CCTV footage. Minor accidents exhibit slight contact between vehicles with minimal

deformation and intact car structures. Substantial accidents show visible damage including bent frames and broken glass. Critical accidents involve complete vehicle destruction with exposed engines and multiple heavily damaged vehicles. These differences are large enough that a MobileNetV2 model pre-trained on ImageNet can distinguish them with high confidence.

Second, the dataset's origin from surveillance video frames results in high intra-class similarity, which is an expected characteristic of CCTV-based datasets rather than a data leakage problem. While consecutive frames from the same video are nearly identical visually, the stratified random split ensures fairness at the file level. The training curves (Fig. 5) demonstrate healthy convergence: both training and validation loss decrease smoothly together, with no divergence indicating overfitting. Validation loss reaches near-zero by epoch 2 and remains stable, while training and validation accuracy track closely throughout training.

Third, external validation was performed on 60 completely new images that were never used during training or validation. The model correctly classified all 60 images (20 per class), maintaining 100% accuracy on truly unseen data. This confirms that the model has learned generalizable features rather than memorizing the training set.

6.4 Real-World Validation

To validate the system beyond the CCTV dataset, a real-world Egyptian road accident image was tested. This image was captured from ground-level (not CCTV angle), contained emergency vehicles (ambulances), bystanders, and a heavily damaged vehicle—conditions completely different from the training data. The system correctly detected the accident with 99.9% confidence and classified it as Critical Impact with 67.4% confidence. YOLOv8 detected 5 vehicles including 2 ambulances (classified as trucks). The lower severity confidence (67.4% vs. 99%+ on CCTV images) is expected given the significant difference in camera angle and distance from training conditions. Nevertheless, the system correctly identified both the presence and critical nature of the accident, demonstrating robustness to domain shift.

7 Complete System Pipeline

When a new image is provided to the system, the following processing steps are executed sequentially as summarized in Table 6.

Table 6. Complete system processing pipeline.

Step	Action
Step 1: Load Image	Read image with OpenCV, convert BGR to RGB
Step 2: YOLO Detection	YOLOv8 identifies vehicle bounding boxes in the scene
Step 3: Preprocess	Resize to 224×224, normalize pixels to [0,1], expand dimensions to (1,224,224,3)
Step 4: Model 1	Accident Classifier predicts Accident or NonAccident with confidence score
Step 5: Model 2 (if Accident)	Severity Classifier predicts Minor / Substantial / Critical with confidence
Step 6: Visualize	Draw bounding boxes, display result with severity confidence bar chart

8 Conclusion

This paper presented a two-stage deep learning system for real-time accident detection and severity analysis from CCTV surveillance images. The proposed bi-model architecture combines MobileNetV2-based transfer learning for classification with YOLOv8-nano for vehicle detection. The system achieved approximately 97% accuracy for accident detection and 100% accuracy for severity classification on the validation set. Real-world testing on an external Egyptian accident image confirmed the system's ability to generalize beyond training conditions, correctly identifying a critical accident with high confidence.

The lightweight nature of the employed models makes the system suitable for deployment on standard hardware in traffic management centers. Future work will focus on extending the system to process video streams in real-time, integrating additional contextual features such as weather conditions and traffic density, and deploying the system on edge devices for distributed processing.

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